INCH-POUND

MIL-A-70952 (AR)
23 March 1990

MILITARY SPECIFICATION

ALTERNATOR, TURBINE (AIR DRIVEN)

This Specification is approved for use within the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements, quality assurance provisions and packaging of the turbine alternator (T/A) for the M745 Point-Detonating (PD) Mortar Fuze.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-P-116	- Preservation, Methods of
MIL-A-48078	- Ammunition, Standard Quality Assurance Provisions, General Specification for

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document, should be addressed to: Commander U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 1390

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STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-331	- Fuze and Fuze Components, Environmental and Performance Test
MIL-STD-1169	- Packaging, Packing and Marking for Shipment of Inert Ammunition Components
MIL-STD-45662	- Calibration Systems Requirements

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from: Military Specifications and Standards, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19911-5094).

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

US ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

PRODUCT AND PACKAGING DRAWINGS

11927121	- Turbine Alternator, Air Driven

INSPECTION EQUIPMENT DRAWINGS

11927112	- Test Equipment
11927113	- Test Vehicle

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer).

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issue of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of the documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

American Society for Testing and Materials (ASTM)

ASTM A344 Testing of Magnetic Materials

(Copies may be purchased directly from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA. 19103).

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Construction. The device shall be constructed in accordance with applicable drawings, specified herein. Any unit failing to satisfy the requirements specified in (3.3.1) shall be excluded from the lots. Variance from dimensions and characteristics as depicted on Interface Drawing No. 11927121 will not be permitted unless request for such variance states that interface dimensions are involved.

3.1.1 Materials. Materials shall be those specified by the applicable drawings.

3.2 Turbine alternator assemblies. The turbine alternator assemblies and parts shall comply with all requirements specified on drawing (dwg.) 11927121 and with all requirements specified in applicable specifications and standards.

3.3 Performance requirements.

3.3.1 Operating requirements. When air pressures of the indicated magnitudes are directed at the turbine of the turbine alternator, the shaft of the device shall rotate at a rate between the indicated minimum and maximum values stated below.

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3.3.1.1 Low pressure requirements. When subjected to an input air pressure of 0.206 pounds per square inch (psi) the shaft rotational speed shall remain within the stated limits over a period of ten seconds.

		Revolutions per minute (rpm)	
	Temperature	Minimum	Maximum
3.3.1.1.1	Ambient	25,000	29,000
3.3.1.1.2	-50 degrees F	20,000	27,000
3.3.1.1.3	+145 degrees F	25,000	30,000

3.3.1.2 High pressure requirements. When subjected to an input air pressure of 3.5 psi for a period of one minute the shaft rotational speed shall not exceed 118,000 rpm. This requirement applies at all temperatures from -50°F to + 145° F. Although a gradual increase or decrease in rpm is permissible during the one minute of operation, there shall be no sudden decrease or increase in rotational rate.

3.3.2 Environmental requirements. The device assembled into a fixture similar to an inert M745 fuze and subjected to the following environmental conditioning will continue to satisfy the requirements of 3.3.1 following that conditioning.

3.3.2.1 Temperature and humidity testing. MIL-STD-331, procedure 105.

3.3.2.2 Other environments. The device is expected to survive all the other non-destructive environments defined in the specification for the M745 PD fuze; e.g., transportation-vibration, rough handling, etc. However, these environments are not specifically evaluated at the device level.

3.4 Workmanship. Parts and assemblies shall be free from chips, dirt, grease and other foreign matter that may impair proper functioning. The cleaning method used shall not be injurious to any of the parts, nor shall the parts be contaminated.

3.5 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Alternate inspection provisions. Alternative inspection procedures, methods or equipment (including tool inspection, statistical process control, substitute sampling plans, etc.) may be used by the contractor when they provide, as a minimum the level of quality assurance required by the provisions specified herein. Prior to applying such alternative inspection procedures, methods or equipment, the contractor shall describe them in a written proposal submitted to the Procuring Contracting Officer for evaluation and approval by the Government. When required, the contractor shall demonstrate that the effectiveness of the proposed alternative(s) is equal to or better than the quality assurance provisions herein. In cases of dispute as to whether certain procedures of the contractor's proposed alternative(s) provide equal assurance, the provisions of this specification shall apply.

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4.2 Classification of inspections. The following types of inspections shall be conducted on this item.

a. First article inspection

b. Quality conformance inspection

4.3. First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with provisions of 4.3.2. The FAAS shall comprise (1) ten sets of the seven piece parts together with measurements of each and every quantified parameter appearing on their respective drawings, (2) 50 sets of the 12 piece parts, subassemblies and final assembly together with the measurements made on them to the Classification of Defects called out in 4.4.2.1 to 4.4.2.12 and (3) 244 turbine alternators subjected to the testing outlined in Figure 1. Fifty units, as comparison samples, selected at random from each accepted lot shall be shipped within two working days of lot acceptance to the technical agency designated by the procuring activity (6.3). If any part is manufactured by a multicavity process; e.g., by stamping, molding, etc., ten properly marked samples from each cavity are required for inspection.

4.3.2 Inspections to be performed. See MIL-A-48078 and figure I specified herein.

4.3.3 Rejection. See MIL-A-48078.

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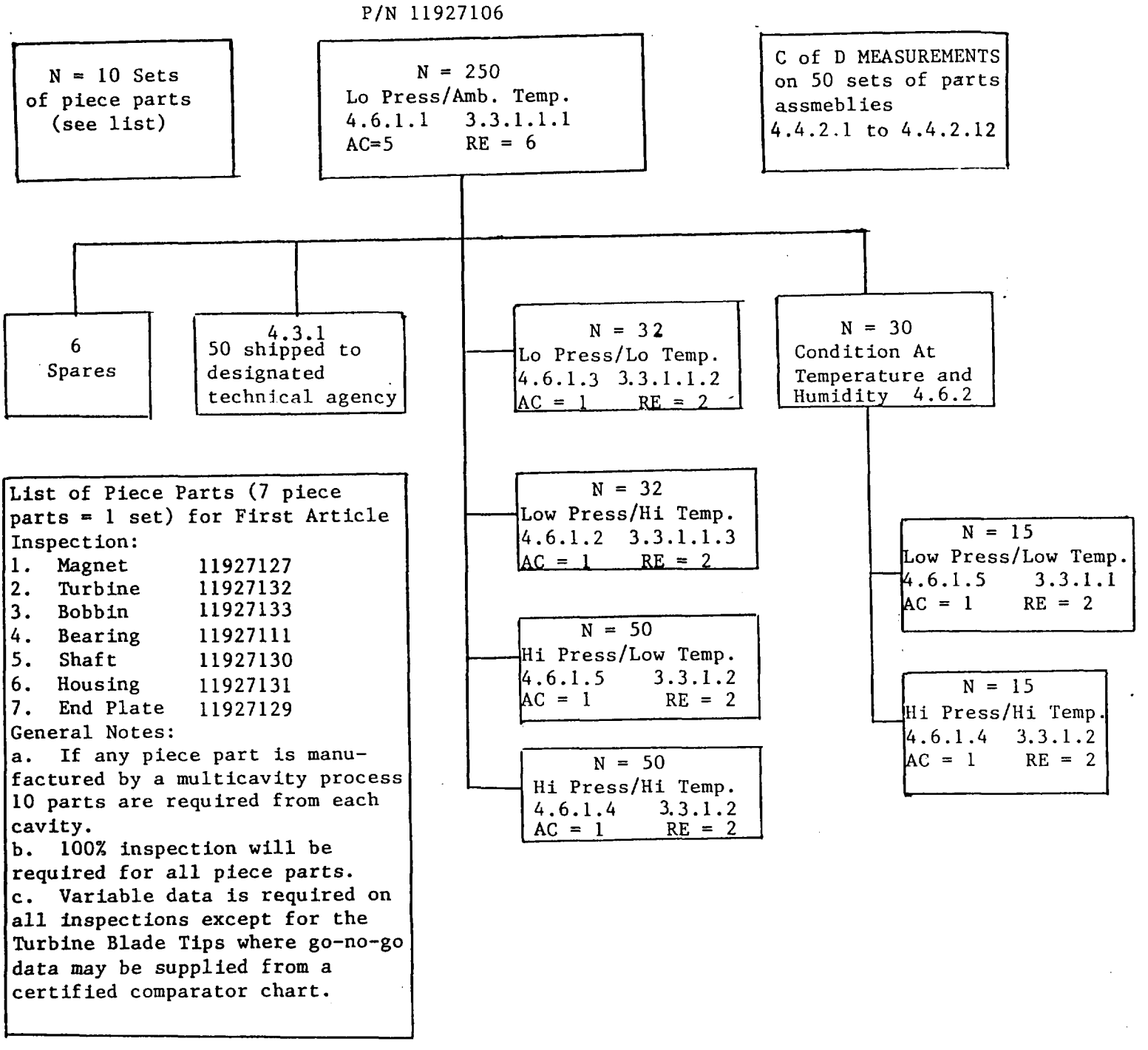


Figure 1 First Article Inspection Flow Diagram

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4.4 Quality Conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

4.4.2 Examination. See MIL-A-48078

4.4.2.1 Sampling plans. Unless otherwise specified in the classification of Defects and Test tables, sampling plans and procedures for major and minor defects shall be in accordance with MIL-STD-105, inspection level II.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Alternator, Turbine (Air Driven)	1 OF 1	11927121
			NEXT HIGHER ASSEMBLY: 11927106

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Exceeds the 0.358 dimension tolerance		0.40%	3.2	Gage
102	Improper shaft end play (0.003 to 0.013)		0.40%	3.2	Gage
103	Improper assembly of turbine to shaft (this is a destructive test)		0.40%	1/	Gage
104	Exceeds the 0.305 dimension tolerance		0.40%	3.2	Gage
105	Turbine on shaft magnet assembly shall rotate within the housing and end plate, with only magnetic coupling resistance, free of physical rubbing contact friction. Test shall be conducted twice with each end of the assembly vertically upward.		0.40%	3.2	Visual/Audible
106	Exceeds the 0.152 dimension tolerance		0.40%	3.2	Gage
107	Improper assembly of end plate to housing		0.40%	3.2	Visual/Gage
108	Weld above housing edge		0.40%	3.2	Visual/Gage
Minor	None defined				

NOTES: 1. Hold turbine and push down on turbine end of shaft with an axial force of 10 pounds. The turbine shall not separate from the shaft.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Magnet-Shaft and Washer Assembly	1 OF 1	11927126
			NEXT HIGHER ASSEMBLY 11927121

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Exceeds 0.002 run out tolerance of outer diameter		0.40%	3.2	Gage
102	Exceeds 0.003 run out tolerance of magnet face		0.40%	3.2	Gage
103	Exceeds 0.043 height tolerance		0.40%	3.2	Gage
Minor					
201	Exceeds 0.073 height tolerance		0.65%	3.2	Gage
202	Surface defects on raceways and datum		0.65%	3.2	Gage
203	Improper adherence of shaft to magnet		0.65%	2/	Gage

NOTES: 2. Hold magnet stationary, push shaft in either direction with an axial force of 120 pounds for a minimum of 10 (ten) seconds. The magnet shall not move or separate from the shaft.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH 4.4.2.3	TITLE Magnet		SHEET 1 OF 1		DRAWING NUMBER 11927127
					NEXT HIGHER ASSEMBLY 11927126
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Structural defects		0.40%	3/	Gage
102	Exceeds 0.6000 outside diameter dimension		0.40%	3.2	Gage
103	Exceeds 0.0005 perpendicularity tolerance of outside surface		0.40%	3.2	Gage
104	Exceeds 0.0005 parallelism tolerance of end surface		0.40%	3.2	Gage
Minor					
201	Exceeds 0.005 true position tolerance of 0.293 diameter		0.65%	3.2	Gage
202	Exceeds 0.005 true position tolerance of 0.237 diameter		0.65%	3.2	Gage
203	Exceeds 0.005 true position tolerance of 0.174 diameter		0.65%	3.2	Gage

NOTES: 3. Spin magnet at a speed of 140,000 rpm for a minimum of five (5) seconds, while being held in an approved shaft-like fixture. The magnet shall not exhibit any structural defects.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-A-70952 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.4	Shaft	1 OF 1	11927130
			NEXT HIGHER ASSEMBLY 11927126

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Improper 16 micro finish on the suraface where the bearing balls make contact, 2 places		0.40%	4/	Visual/Optical
102	Exceeds 0.148 location tolerance		0.40%	3.2	Gage
103	Exceeds 0.325 or (0.349 alternate) location tolerance		0.40%	3.2	Gage
104	Exceeds either 0.1020 diameter tolerance		0.40%	3.2	Gage
105	Exceeds either 0.1105 diameter tolerance		0.40%	3.2	Gage
106	Exceeds the 37° Vee groove angle tolerance		0.40%	3.2	Gage
Minor					
201	Exceeds 0.081 width		0.65%	3.2	Gage

NOTES:

4. The surface of the bearing radii, which forms the bearing interraces, shall be free of any flats, pits, bumps, tool marks or any other surface irregularities, in excess of the specified finish.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-A-70952 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.5	Housing and Bearing Assembly				11927128
					NEXT HIGHER ASSEMBLY 11927121
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Lack of freedom of the ball complement within the raceway		100%	5/	Needle/Gage
102	Exceeds 0.1022 diameter tolerance		0.40%	3.2	Gage
103	Exceeds 0.2055 outside diameter tolerance of bearing boss		0.40%	3.2	Gage
Minor	None defined				

NOTES: 5. Carefully insert the needle gage between two balls. Gently drive the ball complement for at least 90° of rotation, all 13 balls shall rotate in the raceway.

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MIL-A-70952 (AR)

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.6	Housing	1 OF 1	11927131
			NEXT HIGHER ASSEMBLY 11927128

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Exceeds 0.1743 diameter tolerance		0.40%	3.2	Gage
102	Exceeds 0.2055 diameter tolerance		0.40%	3.2	Gage
103	Exceeds 0.355 height tolerance		0.40%	3.2	Gage
104	Exceeds 0.0685 height tolerance		0.40%	3.2	Gage
105	Exceeds 1.128 inner diameter and true positional tolerance		0.40%	3.2	Gage
Minor					
201	Exceeds 0.625 diameter tolerance		0.65%	3.2	Gage
202	Incorrect max material conditions of housing base cut-outs		0.65%	3.2	Gage
203	Exceeds 0.056 height tolerance		0.65%	3.2	Gage

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.7	Bearing	1 OF 1	11927111
			NEXT HIGHER ASSEMBLY: 11927110

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Exceeds 0.1022 inner ball circle diameter tolerance		0.40%	3.2	Gage
102	Incorrect numbers of balls		0.40%	3.2	Visual
103	Exceeds 0.1748 outside diameter tolerance		0.40%	3.2	Gage
Minor					
201	Exceeds 0.115 inner diameter tolerance		0.65%	3.2	Gage
202	Exceeds 0.1065 inner diameter tolerance		0.65%	3.2	Gage

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-A-70952 (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER 11927078
4.4.2.8	Lubricant Mixture, Bearing		NEXT HIGHER ASSEMBLY 11927111

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	As-mixed ratio out-of-tolerance		once/ batch	6/	Gage
102	Ratio in usage above maximum ratio		once/lubricated bearing lot	7/	Gage
Minor	None defined				

NOTES: 6. Weigh sample of mixture in an approved evaporation dish of known weight. Depth of solution shall not exceed 0.010 inch. Place in oven set at 180°F + 10°F for two hours. Reweigh dish and contents. The weight of the residue shall be 9.52% to 10.53% of the original weight of the mixture. Alternative methods must be approved by the government.

7. Draw a sample of the mixture after bearing lubrication. A sample must be drawn with each lot of bearings lubricated. Weight sample and place in approved evaporation dish of known weight to a depth of no greater than .010 inch. Place in oven set at 180°F + 10°F for two hours. Weight of the residue shall be between 9.52% and 12.5% of the original weight. Alternative methods must be approved by the government.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.4.2.9	End Plate and Bearing Assembly	1	1	11927110
				NEXT HIGHER ASSEMBLY: 11927121

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Lack of freedom of the ball complement within the raceway		100%	5/	Needle/Gage
102	Exceeds 0.1022 inner ball circle diameter tolerance		0.40%	3.2	Gage
Minor	None defined				

NOTES:

5. Carefully insert the needle gage between two balls. Gently drive the ball complement for at least 90° of rotation, all 13 balls shall rotate in the raceway.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-A-70952 (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.10	End Plate				11927129
					NEXT HIGHER ASSEMBLY 11927110
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Exceeds 0.1743 inner daimeter tolerance		0.40%	3.2	Gage
102	Exceeds 1.130 outer diameter tolerance		0.40%	3.2	Gage
103	Exceeds 0.047 height tolerance		0.40%	3.2	Gage
Minor					
201	Exceeds 0.125 associated true position tolerance of the .125 diameter		0.65%	3.2	Gage
202	Exceeds 0.625 diameter tolerance		0.65%	3.2	Gage
203	Exceeds max material condition of 0.280 slot, 3 places		0.65%	3.2	Gage

NOTES:

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.11	Turbine, 10-Blade	1 OF 1	11927132
			NEXT HIGHER ASSEMBLY: 11927121

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Exceeds 0.082 diameter tolerance		0.40%	3.2	Gage
Minor					
201	Exceeds tolerance of 0.758 diameter		0.65%	3.2	Gage
202	Exceeds tolerance of 0.145 height dimension		0.65%	3.2	Gage

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.12	Bobbin	1 of 1	11927133
			NEXT HIGHER ASSEMBLY 11927121

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Incorrect location of the 0.060 diameter bosses per Table		0.40%	3.2	Gage
102	Exceeds 0.438 location tolerance of four tips		0.40%	3.2	Gage
103	Exceeds 0.0625 location tolerance four places		0.40%	3.2	Gage
Minor					
201	Exceeds 0.283 height tolerance		0.65%	3.2	Gage
202	Exceeds 0.060 diameter tolerance of three bosses		0.65%	3.2	Gage
203	Exceeds 0.704 inner diameter tolerance		0.65%	3.2	Gage

NOTES:

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4.4.3 Lot acceptance test. The lot acceptance test sample shall be subjected to the test group, requirements and procedures specified below. All tests shall be performed by the contractor and witnessed by the appropriate government personnel.The turbine alternators shall be randomly sampled for lot acceptance testing in accordance with the specified level of MIL-STD-105.

GROUP	TEST REQUIREMENT	TEST PROCEDURE	AQL(%)
		TEST	
1	3.3.1.1 (low pressure -50°F)	4.6.1 & 4.6.1.3	Level S4 1.5%
2	3.3.1.2 (Hi pressure +145°F)	4.6.1 & 4.6.1.4	Level 1 1.0%

4.4.3.1 Disposition of lot acceptance sample. The samples tested in (4.4.3) shall not be returned to the lot. The nondestructive samples (low pressure) are to be stored at the contractor's plant for a period not to exceed one (1) year. Disposition to be made by the procuring agency.

4.4.4 Material and component certification. Prior to acceptance under this procurement, the contractor shall demonstrate by means of certifications or statements of findings that only materials and components conforming to the item specification and drawings have been used.

4.5 Test conditions and equipment.

4.5.1 Test conditions. Unless otherwise specified, all measurements are to be made at normal room conditions of temperature, relative humidity and altitude. Temperature conversion between Fahrenheit and Celsius shall be carried to the nearest whole degree. The following table of tolerance shall be applied where applicable to specified test conditions or measurements, not to the equipment used for these measurements unless otherwise defined in the operating requirements.

a.	Temperature	± 2° F
b.	Force, weight, or torque	± 5%
c.	Rotational speed (rpm)	± 2%
d.	Pressure	- 3%

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4.5.2 Testing. See 4.6

4.5.3 Inspection equipment. The inspection equipment required to perform the examinations and test prescribed herein is described in the "Paragraph Reference/Inspection Mehtod" column in the tables starting with paragraph 4.4.2.1 and 4.4.3. Calibration procedures shall conform to MIL-STD-45662. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.4 herein.

4.5.3.1 Alternator acceptance test set. The alternator acceptance tests shall be performed on an approved acceptance test set (Dwg 11927112 or an approved equivalent). The design of this acceptance tester shall be such as to control the specified air pressure with traceability to an approved calibration procedure. This tester as a minimum shall accommodate the alternator test vehicle (Dwg 11927113) and provide means of measuring pressure, rpm, while recording rpm. The rpm shall be recorded 15 seconds after achieving the proper air pressure.

4.6 Methods of inspection.

4.6.1 General inspection procedures. The alternator shall be mounted in the test set-up (Dwg 11927112) or any other government approved test set-up, as per paragraph (4.5.3.1), and subjected to the pressure and temperature conditions specified below. When a temperature condition other than ambient is required the alternator shall be conditioned at the required temperature for at least two hours. The test shall then be initiated within fifteen seconds after the alternator has been removed from the conditioning chamber.

4.6.1.1 Low pressure ambient test. The alternator shall be tested in accordance with 4.6.1 and shall meet the ambient requirements of 3.3.1.1. The one minute time does not apply.

4.6.1.2 Low pressure high temperature. The alternator shall be tested at +145° F in accordance with 4.6.1 and shall meet the requirement of 3.3.1.

4.6.1.3 Low pressure low temperature test. The alternator shall be tested at -50° F in accordance with 4.6.1 and shall meet the requirements of 3.3.1.1.

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4.6.1.4 High pressure high temperature test. The alternator shall be tested at 145° F in accordance with 4.6.1 and shall meet the requirements of 3.3.1.2.

4.6.1.5 High pressure low temperature test. The alternator shall be tested at -50° F in accordance with 4.6.1 and shall meet the requirements of 3.3.1.2.

4.6.2 Temperature humidity test. A sample of thirty M745 alternators shall be selected, inserted into dummy M745 fuze bodies or other government approved test vehicles and subjected to the test as specified in MIL-STD-331. procedure 105. The test unit shall then be disassembled and the alternator be tested in accordance with 4.6.1.3 and 4.6.1.4. Any alternator that does not meet the performance requirements of 3.3.1.1 and 3.3.1.2 shall be classified defective (see figure 1).

5. PACKAGING

5.1 Packing. Each alternator shall be packaged in a clean container which shall limit the motion of the device within the container to 0.020 inch maximum in any direction, in accordance with MIL-P-116.

5.2 Marking. In addition to any special marking required by the contract or order, unit packages, intermediate packages, and shipping containers shall be marked in accordance with the requirements of MIL-STD-1169.

6. NOTES

6.1 Intended use. The turbine alternator, air driven, is designed to provide a high reliability source of mechanical power for arming in Fuze, Mortar: PD: M745 for 60mm Light-weight Company Mortar System.

6.2 Acquisition requirements. See MIL-A-48078

6.3 Government testing of comparison samples. The comparison samples required by 4.3.1 and as part of the FAAS are subject to periodic retesting by the government to insure that the first article approval requirements and test equipment certification continue to be met. Failure of these samples will not affect the acceptance of the lot but may indicate future mandatory action.

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6.4 Submission of designs for approval. See 6.2.3 of MIL-A-48078 (AR). Submit equipment designs, as required to: Commander, U.S. Army Armament Research, Development and Engineering Center, ATTN: AMSMC-QAT-I (D), Picatinny Arsenal, 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract. Unless otherwise specified, data item DI-R-1714 will apply.

6.5 Alternate design. The only government approved source for the turbine alternator as specified by MIL-A-70952 (AR) is: Alinabal, Inc., 28 Woodmont Rd, Milford, CT 06460

6.6 Subject term (key word) listing.

Alternator, turbine (air driven)
Fuze, PD: M745

Custodian
Army-AR

Preparing activity:
Army-AR

(Project 1390-A682)

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-A-70952 (AR)	23 March 1990

3. DOCUMENT TITLE

ALTERNATOR, TURBINE (AIR DRIVEN)

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
US Army ARDEC Standardization & Specification Office	(1) Commercial (201) 724-6671	(2) AUTOVON 880-6671
c. ADDRESS (Include Zip Code) ATTN: SMCAR-BAC-S/Bldg 6 Picatinny Arsenal, NJ 07806-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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